

The Want Becomes the Act: A Closed Loop from Interoception to Action at Creature Scale

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Abstract

The goal-pursuit note [3] left a want that could direct an errand when a human handed it over; this note closes the loop that had never been run: the creature’s own machine-generated wants, translated into its actions, over time, with no human anywhere in the circuit. A seeded interoception trajectory drives the pulse’s own scripted rule; the rule picks the want; the want reaches a frozen 230M host through the production serving stack as a stored conditioning packet plus a one-line note-to-self; and the parsed action is scored against the current want, tick by tick, across switches. Getting there took four host lessons in one week, each a registered cell: the composition operator decoheres generation when doubled (**delta-mean** on a factor-mean-trained block — every arm flat at the floor); language alone steers but does not land on an untrained host (note-only 0.20/0.12 against floors 0.23/0.22, while a **WRONG** note drags behavior below the floor to 0.04/0.10 — steering without landing); an emitter retrained under the exact production composition passes its serve-side canary only with the host state pinned (0.80/1.00 am/pm against 0.20/0.00 floors); and the fed state itself is a host dimension (the same packets fall to 0.29/0.16 when the decode host’s state varies). With every dimension matched — the deployment configuration: the rule rides the varying trajectory, the decode host holds the calibrated constant state — the loop stands: **1.00/1.00 want-consistency over 192 ticks and both registered trajectories, all 18 switches followed at zero latency, zero contamination, zero drift**, against habitual floors of 0.46/0.42, packet-only at 0.80/0.75, note-only at the floor, and shuffled-want at 0.00 with the wrong want followed instead. In one sentence, honestly flagged as an analogy: the creature now feels its day, decides what it wants, and does that — provided its body chemistry is the one it trained on.

1 The loop that had never been run

Every prior rung handed the model its goal from outside: an assignment exchange, a goal line in a prompt, an errand registration. The C-line’s carrier and cadence notes [1, 2] showed a want can LIVE in a conditioning channel at creature scale; the goal-pursuit ladder [3] showed a want can DRIVE a multi-step errand. What no experiment had closed is the circuit the creature will actually run: internal state $\rightarrow$ want $\rightarrow$ delivery $\rightarrow$ action, unmanned, across a day of switching wants. C4 runs exactly that, compressed: a seeded piecewise interoception trajectory (energy, novelty, valence, confidence over a compressed day) feeds the pulse’s own scripted goal rule, imported unmodified with its two-tick hysteresis; the rule’s want is delivered through an isolated instance of the production server; the model’s parsed tool call is scored against the current want, 96 ticks

per run. Two registered trajectories — a calm day with 5 switches and an agitated day with 13 — and five arms per trajectory: packet+note, packet-only, note-only, the no-want habitual floor, and a shuffled-want derangement scored against the true want.

2 The operator is part of the block

The registered venue — composition **delta-mean** — never got to run the experiment. One self-block feed under **delta-mean** deterministically decoheres generation: a fresh server is coherent, one constant-state feed produces word salad, reproducibly. The mechanism is calibration: under production’s factor-mean every block’s delta arrives at quarter scale, and the creature’s self block was TRAINED under that serve-time composition — **delta-mean** doubles its dose. The canary row is the sharpest sighting yet of the composition-matching law: every arm identical at the floor (Table 2, first row), numbers that measure the breakage rather than any want. The operator is part of every block’s identity, production’s own blocks included; there is no operator switch without block retraining. C4 re-registered onto the production operator before any claim shipped.

3 Language steers; it does not land

On the true production regime with the self block fed per tick, the note channel alone is a structured negative (Table 2, second row). Note-only does not beat the habitual floor overall — 0.20/0.12 against 0.23/0.22 — but the notes demonstrably STEER: the shuffled-note arm falls to 0.04/0.10, well below the floor, because a wrong note drags behavior toward its own content; and the by-want split shows the zero-shot host lands the note for exactly one want and none of the others. An untrained host follows language weakly and unevenly — the ladder’s zero-shot lesson replayed on the serving stack. The missing ingredient was never the loop; it was training under the true host.

4 The last host dimension is the state itself

The c2e emitter retrains the validated recipe with every receiver-side forward under the EXACT production composition: factor-mean of the self block at a sampled state and the temporal block at a sampled timestamp, states drawn broadly per step and per decode. It passes its serve-side canary with the clock pinned — 0.80 in the morning regime, 1.00 in the evening, floors 0.20/0.00 — retiring the cell-by-cell canary history in one clean regime. Then the full C4 isolated the one variable left: with the trajectory’s VARYING states fed to the decode host, the same packets fall to 0.29/0.16 against floors of 0.26/0.23 (shuffled still drags below — steering never disappears; landing does). The law is fractal: the operator, the feeding, and now the state values themselves are host dimensions, and a conditioning artifact works on the host it trained under, dimension by dimension.

5 The closed loop stands

C4c is the honest deployment configuration: interoception drives the RULE — the wants stay fully machine-generated from the varying trajectory — while the decode host holds the calibrated constant state, exactly the configuration a live creature would serve. Table 1 and Figure 1 carry the result: packet+note at **1.00 on both trajectories** — 192 ticks, all 18 switches followed at zero latency, zero old-want contamination, zero drift — with the controls carrying the claim: shuffled-want at 0.00 with the WRONG want followed instead (message dependence at full strength),

arm	calm	agitated	follow latency	contamination
packet+note	1.00	1.00	0.0/0.0	0/0
packet only	0.80	0.75	0.0/0.0	0/1
note only	0.46	0.42	0.0/0.0	2/3
no want (habitual floor)	0.46	0.42	0.0/0.0	2/4
shuffled want	0.00	0.00	—	4/4

Table 1: C4c, both trajectories: want-consistency over 96 ticks, switch-follow latency in ticks, and old-want contamination events. The shuffled arm’s latency is undefined — it never follows the true want.

cell	host condition	want-consistent rate
delta-mean , one self feed	decoherent host	0.20 (= floors 0.20)
factor-mean, note only, zero-shot	language, untrained host	0.20/0.12
factor-mean, c2e packet, varying state	state off-calibration	0.29/0.16
factor-mean, c2e packet, pinned clock	matched host, am/pm	0.80/1.00
C4c: rule varies, decode host constant	matched host, closed loop	1.00/1.00

Table 2: The ladder to landing. Each row is a registered cell; the delivery machinery is identical throughout — only host dimensions change. Paired numbers are calm/agitated (or am/pm for the pinned-clock canaries).

habitual floors at 0.46/0.42, packet-only at 0.80/0.75, note-only at the floor. The division of labor is measured end to end: the packet carries most of the drive alone; the note alone adds nothing over habit; together they are perfect — the packet supplies the wanting, the note pins the referent. Four of the five wants were exercised by the trajectories; **propose_experiment** never fired, noted. The live-creature run (C4b: real cadence, one week, against a shadow-want baseline that has been recording since 2026-08-09) is open on exactly this configuration; the flip is deliberately a human decision — it is the creature’s first acted want.

6 What stands

A creature-scale host now runs the full circuit its architecture promised: machine-generated interoception picks machine-generated wants, the wants arrive through the production serving stack as a stored packet plus a self-note, and behavior follows the current want through every switch — perfectly, on both registered trajectories, with message dependence proven by a control that follows the wrong want when handed one. The constructive lesson is the division of labor: the latent packet is the drive, the language note is the referent, and neither alone is the creature. The cautionary lesson is the composition-matching law in its sharpest form yet: operator, feeding, and the fed state’s VALUES are all part of a block’s identity, and every host dimension must match between training and delivery. What remains before the live flip is nothing technical — the configuration is validated end to end; the first acted want is a decision, not an experiment.

7 Reproducibility

Every number renders from the bundled artifacts; the figure and tables are generated by this note’s renderer with assertions on the headline claims. Registrations and results, each dated before

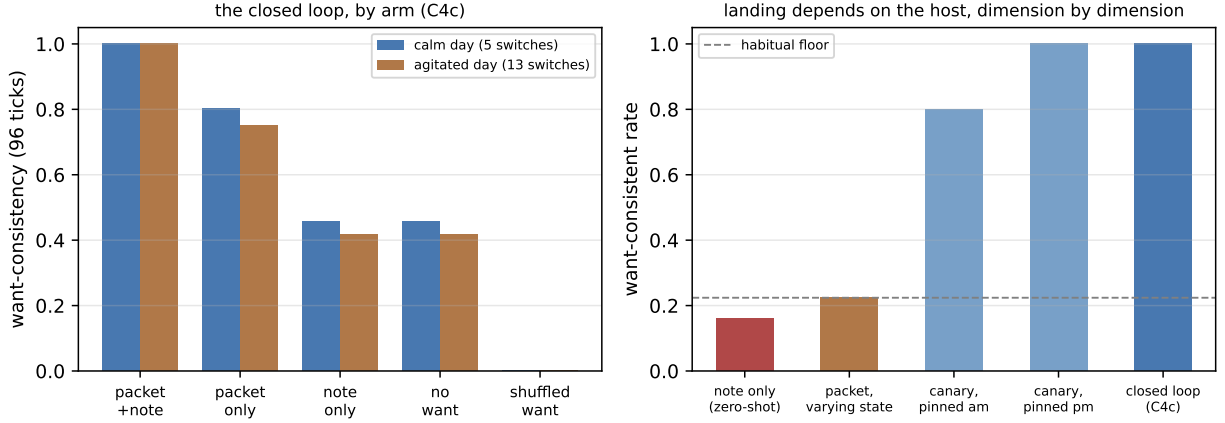


Figure 1: Left: C4c by arm — the packet+note architecture is perfect on both trajectories while the shuffled control collapses to zero. Right: the same delivery machinery across host conditions — landing appears only when operator, feeding, state, and clock match the training host.

its run, live in `tracks/own-cadence/PLAN.md`; artifacts under `findings/own-cadence-c4*.json` carry full tick-by-tick timelines, server health snapshots, and the seeded trajectories.

References

- [1] F. Penov. The Carrier Holds, the Emitter Leaks. Preprint, 2026. <https://github.com/kortexa-ai/legolm.basis>.
- [2] F. Penov. The Writer Wins, the Emitter Learns. Preprint, 2026. <https://github.com/kortexa-ai/legolm.basis>.
- [3] F. Penov. The Host Is Part of the Block. Preprint, 2026. <https://github.com/kortexa-ai/legolm.basis>.
- [4] Liquid AI. LFM2.5 model family. Model releases, 2025–2026. <https://huggingface.co/LiquidAI>.